```
        path: /etc/apt/sources.list.d/elastic-5.x.list
        create: yes
        line: 'deb https://artifacts.elastic.co/packages/5.x/
apt stable main'

    - name: Update the software package repository
      apt:
        update_cache: yes

    - name: Install Elasticsearch
      package:
        name: elasticsearch
        state: latest

    - name: Update cluster name
      lineinfile:
        path: /etc/elasticsearch/elastisearch.yml
        create: yes
        regexp: '^#cluster.name: my-application'
        line: 'cluster.name: graylog'

    - name: Daemon reload
      systemd: daemon_reload=yes

    - name: Start elasticsearch service
      service:
        name: elasticsearch.service
        state: started

    - wait_for:
        port: 9200

    - name: Test Curl query
      shell: curl -XGET 'localhost:9200/?pretty'
```

The stable *elastic.co* repository package is installed before installing Elasticsearch. The cluster name is then updated in the */etc/elasticsearch/elasticsearch.yml* configuration file. The system daemon services are reloaded, and the Elasticsearch service is started. The Ansible playbook waits for the service to run and listen on port 9200.

The above playbook can be invoked as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/graylog.yml --tags elastic -K
```

You can perform a manual query to verify that Elasticsearch is running using the following Curl command:

```
$ curl -XGET 'localhost:9200/?pretty'

{
  "name" : "cFn-3YD",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "nuBTSlFBTk6PDGyrfDCr3A",
  "version" : {
    "number" : "5.6.5",
    "build_hash" : "6a37571",
    "build_date" : "2017-12-04T07:50:10.466Z",
    "build_snapshot" : false,
    "lucene_version" : "6.6.1"
  },
  "tagline" : "You Know, for Search"
}
```

Graylog

The final step is to install Graylog itself. The *.deb* package available from the *graylog2.org* website is installed and then the actual 'graylog-server' package is installed. The configuration file is updated with credentials for the 'admin' user with a hashed string for the password 'osfy'. The Web interface is also enabled with the default IP address of the guest VM. The Graylog service is finally started. The Ansible playbook to install Graylog is as follows:

```
- name: Install Graylog
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [graylog]

  tasks:
    - name: Install Graylog repo deb
      apt:
        deb: https://packages.graylog2.org/repo/packages/
graylog-2.3-repository_latest.deb

    - name: Update the software package repository
      apt:
        update_cache: yes

    - name: Install Graylog
      package:
        name: graylog-server
        state: latest

    - name: Update database credentials in the file
      replace:
        dest: "/etc/graylog/server/server.conf"
        regexp: "{{ item.regexp }}"
        replace: "{{ item.replace }}"
      with_items:
        - { regexp: 'password_secret =', replace: 'password_
secret = QXHg3Eqvsu PmFxUY2aKlgimUF05plMPXQ Hy1stUiQ1uaxgIG27
K3t2MviRiFLNot09U1ako T30njK3G69KIzqIoYqdY3oLUP' }
        - { regexp: '#root_username = admin', replace: 'root_
username = admin' }
```